



Fig. 7: Switch-mode charge cycles with and without PS700, 2200mAh

Typically, integration encompasses charge-current sensing, the pass transistor and reverse-discharge protection. In addition, these charge-management controllers typically employ some type of thermal regulation. Thermal regulation optimises the charge cycle time while maintaining device reliability by limiting the charge current based on the device die temperature. Thermal regulation greatly reduces the thermal design effort.

Fig. 4 depicts a fully integrated, linear solution utilising the MCP73861. The MCP73861 incorporates all the features of the MCP73843 along with charge-current sensing, the pass transistor, reverse-discharge protection and cell temperature monitoring.

Charge-cycle waveforms. Fig. 5 depicts complete charge cycles utilising the MCP73843 with constant-current charge rates of 1C and 0.5C. Charging at a rate of 0.5C instead of 1C takes about one hour longer for the end of charge to be reached. The MCP73843 scales the charge termination current proportionately with the fast charge current. The result is an increase of 36 per cent in charge time with the benefit of a 2 per cent gain in capacity and reduced power dissipation. The change in termination current from 0.07C to 0.035C results in an increase in final capacity from around 98 per cent to 100 per cent. The system

designer has to make a trade-off between charge time, power dissipation and available capacity.

Switch-mode charging solutions.

Switch-mode charging solutions are generally employed in applications that have a wide-ranging input or a high input-to-output voltage differential. In these applications, switch-mode solutions have the advantage of improved efficiency. The disadvantages are system complexity, size and cost.

Take, for example, an application required to charge a 2200mAh, single Li-ion cell from a car adaptor at a constant-current charge rate of 0.5C or 1C. It would be extremely difficult to utilise a linear solution in this application due to the thermal issues involved. A linear solution employing thermal regulation could be utilised, but the charge cycle times at the reduced charge currents may be prohibitive.

The first step to designing a successful switch-mode charging solution is to choose a topology: buck, boost, buck-boost, flyback, single-ended primary inductive converter (SEPIC) or other. Knowing the input and output requirements, and experience, quickly narrows the choices down to two for this application: buck or SEPIC. A buck converter has the advantage of requiring a single inductor. Disadvantages of this topology include an additional diode required for reverse-discharge

protection, high-side gate drive and current sense, and pulsed input current (EMI concern).

The SEPIC topology has advantages that include low-side gate drive and current sense, continuous input current and DC isolation from input to output. The main disadvantage of the SEPIC topology is the use of two inductors and an energy transfer capacitor.

Fig. 6 depicts a schematic for a switch-mode charger. A high-speed pulse-width modulator (PWM), MCP1630, has been utilised in a pseudo smart battery charger application. The MCP1630 is a high-speed, microcontroller-adaptable, pulse-width modulator. When used in conjunction with a microcontroller, the MCP1630 will control the power system duty cycle to provide output voltage or current regulation. The microcontroller, PIC16F684, can be used to regulate the output voltage or current, switching frequency and maximum duty cycle. The MCP1630 generates duty cycle, and provides fast over-current protection based on various external inputs. External signals include the input oscillator, the reference voltage, the feedback voltage and the current sense. The output signal is a square-wave pulse. The power train used for the charger is SEPIC.

The microcontroller provides an enormous amount of design flexibility. In addition, it can communicate with a battery monitor inside the battery pack to significantly reduce charge-cycle times.

Charge-cycle waveforms. Fig. 7 shows complete charge cycles utilising the switch-mode charging solution. By utilising a battery monitor in the charging system, charge cycles can be significantly reduced. The battery monitor eliminates sensing the voltage produced across the pack's protection circuitry and contact resistance by the charging current. ●

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